## 9.3 The Integral Test

Corollary of Monotonic Sequence Theorem:

Let $\{a_n\}$ be a sequence of nonnegative real numbers. Then the series $\sum a_n$ converges if and only if the sequences $\{s_k\}$ of partial sums is bounded.

Example: The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

# The Integral Test

Example: The 2-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

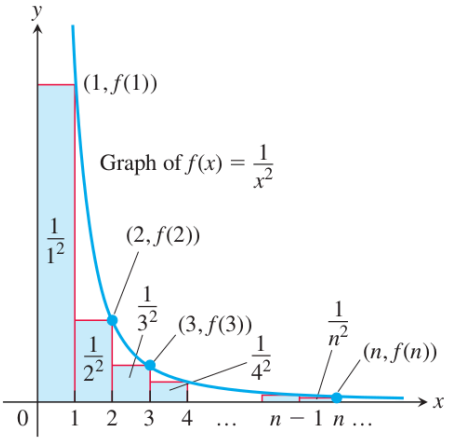
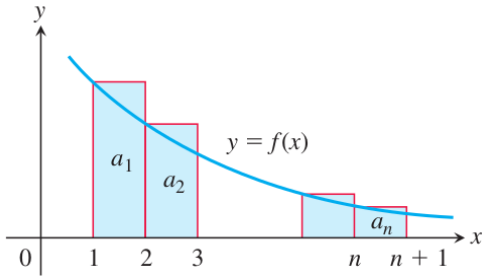
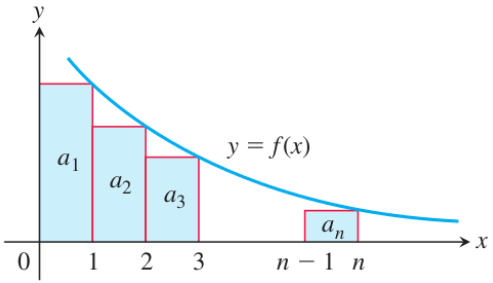



**FIGURE 9.11** The sum of the areas of the rectangles under the graph of $f(x) = 1/x^2$ is less than the area under the graph (Example 2).

Theorem (The Integral Test): Let $\{a_n\}$ be a sequence of positive real numbers. Suppose that $a_n = f(n)$, where $f$ is a continuous, positive, decreasing function of $x$ for all $x \geq N$. Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x)\,dx$ both converge or both diverge.



Example: Given a real constant $p$, the $p$-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Example: The series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ is convergent.

Example: The series $\sum_{n=2}^{\infty} \frac{1}{n\ln n}$ is divergent.

**Error Estimation**

Suppose the series $\sum_{n=1}^{\infty} a_n$ converges to $S$, we want to estimate the size of the **remainder** $R_n$ between $S$and its $n$th partial sum $s_n$. That is,

$$R_n = S - s_n = a_{n+1} + a_{n+2} + a_{n+3} + \ldots$$

**Bound for The Remainder in the Integral Test**

Suppose $\{a_k\}$ is a sequence of positive numbers with $a_k = f(k)$, where $f$ is a continuous positive decreasing function of $x$ for all $x \geq n$, and that $\sum a_n$ converges to $S$. Then the remainder $R_n = S - s_n$ satisfies the inequalities

$$\int_{n+1}^{\infty} f(x)\,dx \leq R_n \leq \int_{n}^{\infty} f(x)\,dx.$$

Example: Estimate the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ using the above inequalities with $n = 10$.

Extra Examples:

1. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(\sqrt{n}+1)}$
2. $\sum_{n=1}^{\infty} \frac{e^{n}}{1+e^{2n}}$
3. $\sum_{n=1}^{\infty} ne^{-n^{2}}$